03 — FIELD PROJECT · 1 OF 2

Accessibility Ramp Systems

Eight modular aluminum wheelchair ramps, configured to code and installed at clients' homes.



FIG. 3.1 — Finished straight run, Mar 2026

CONTEXT	ROLE	PERIOD	METHODS
Action Medical Home Health	Junior Mobility Systems Technician	2025 – 26	Site survey · slope & run · modular layout · footings · install

8	12–26 ft	1 : 13	~1 day
RAMP SYSTEMS CONFIGURED	RANGE OF SPANS	ACHIEVED SLOPE, INSIDE THE 1 : 12 LIMIT	OF WORK PER SYSTEM, START TO FINISH

01 BRIEF

Someone who uses a wheelchair or walker needs to get from the driveway to their front door without help. For many clients a ramp is the difference between leaving home on their own and not leaving at all.

The design space is narrow. Slope is capped at 1 : 12, the rise is fixed by the height of the threshold, and the footprint is whatever the property leaves once the driveway and existing steps are accounted for. The structure has to tie into what is already there (a deck, a set of steps, a porch), carry wheelchair, walker and caregiver traffic every day, and hold its geometry through Ontario freeze–thaw cycles. Every house resets the problem, and the person using the ramp can't wait for a second attempt.

02 CONSTRAINTS

SLOPE	1 : 12 maximum: 1 in of rise per 12 in of run
RISE	Measured grade to door threshold, every site
FOOTPRINT	Whatever the property allows, without blocking the driveway or access
INTERFACES	Existing decks and steps; gravel, soft ground and pavement
PARTS	Standard modules only: sections, turn platforms, landings, legs, handrails
USERS	Wheelchairs, walkers and caregivers, every day, in all weather

03 THE DESIGN PROBLEM

Slope fixes length. Run is the rise multiplied by the slope ratio, so at the 1 : 12 limit a 36 in rise demands 432 in, or **36 ft**, of sloped deck before a single landing is added. The property has no say in that number; the only design freedom is in how the run is arranged in plan.

On the August 2025 switchback, the side yard couldn't hold anything near 36 ft in a straight line. Folding the run back on itself, two parallel legs joined by a full-width turn platform, converts a length problem into a width problem, and width was the one dimension the site could spare.

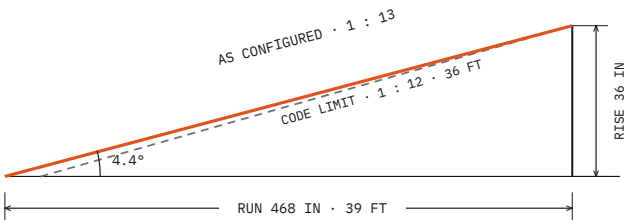


FIG. 3.2 — Same rise, two slopes. Going gentler than the limit costs 3 ft of run.

Min. run at 1 : 12	36 in × 12 = 432 in = 36.0 ft
Run at 1 : 13	36 in × 13 = 468 in = 39.0 ft
Gradient	7.7 % vs 8.3 % limit
Angle from horizontal	atan(1/13) = 4.40°

Why go gentler than the limit? 1 : 12 is a maximum, not a target. Every bit of slope under it makes a ramp easier to climb under your own power and safer to come down, so when the property allows the extra run, the gentler layout wins.

04 PROCESS

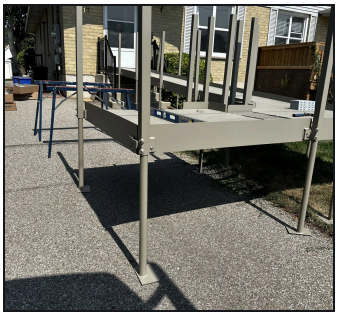
01 Survey the site
Measured the vertical rise from grade to the door threshold and walked the usable ground around the entrance: where the steps come down, how much driveway has to stay clear, what the ground is made of.

02 Configure the layout
Broke the required run into standard modules (straight sections, turn platforms and landings), then checked the slope the layout actually achieves against the slope selected on the layout sheet, and rearranged until it fit the property without crowding the driveway or blocking access.

03 Plan the fabrication
Turned each layout into a parts list and a staging plan, so modules, legs and handrails came off the truck in the order they went in. On the switchback: M1 × 9, M3 × 2, M4 × 1, M6 × 1, M7 × 2.

04 Set footings and level
Coordinated structural footing placement against what was already on the property: decks, steps, soft ground and gravel. Set the adjustable legs and levelled the frame before any decking went on.

05 Install
Installed the perforated deck sections, turn platforms and continuous handrails on both sides, so there is something to hold the full length of the ramp.



3.4 Frame and legs, switchback



3.5 Posts and landing frame, Mar 2026

Not enough yard	FAILURE MODE 3.A
OBSERVED	A 36 in rise needs at least 36 ft of ramp at code slope. The side yard couldn't hold anything close to that in a straight line.
CONSTRAINT	Slope can't be steepened to save length. Run is fixed by the rise and the code.
RESOLVED	Fold it. Two parallel legs joined by turn platforms (module M3) pack about 39 ft of ramp into a 25.5 × 9.8 ft footprint.
VERIFIED	Layout sheet: actual slope 1 : 13 (≈ 4.4°), gentler than the 1 : 12 limit.

Ground that isn't level	FAILURE MODE 3.B
OBSERVED	Every footing sits at a different grade: gravel, soft ground, pavement, or the edge of an existing deck.
RISK	A ramp that follows the ground twists and changes slope along its length.
RESOLVED	Set each adjustable leg individually so the deck plane stays true from bottom to top, regardless of the ground under it.

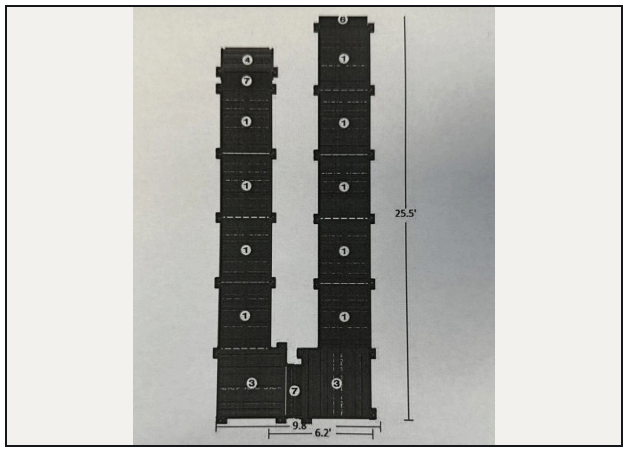


FIG. 3.3 — Layout sheet, switchback install (Aug 2025)

05 OUTCOME

Clients got step-free access to their homes on ramps at or below the code slope limit. Eight systems spanning 12 to 26 ft, each close to a full day's work, from a tight switchback beside a house in August 2025 to a long straight run with a landing at a rural home in March 2026.

06 WHAT I TOOK AWAY

Engineering is judged by whether it works for the person using it, not by how it looks on the layout sheet. The numbers decide what's possible; the site and the client decide what's right.

07 REV. B

- Model the site and module layout in CAD before the first visit, so clearances and footing points are checked before anything is unloaded.
- Survey ground slope at every footing on the measuring visit, so leg heights can be preset.
- Turn the slope and run arithmetic into a calculator. A first version is on my portfolio site.